This assignment will not be graded and is only for practice.

Level 1

1) Consider the following statements:

Statement 1: $0(0,0) + 0(1,1) = (0,0)$ implies that the set $\{(0,0),(1,1)\}$ is a linearly independent subset of $\mathbb{R}^2$.
Statement 2: $2(1,0) + 2(0,1) = (2,2)$ implies that the set $\{(1,0),(0,1),(2,2)\}$ is a linearly dependent subset of $\mathbb{R}^2$.
Statement 3: $2(1,0) + 2(0,1) = (2,2)$ implies that the set $\{(1,0),(2,2)\}$ is a linearly dependent subset of $\mathbb{R}^2$.

Find the number of correct statements.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

2) If $\alpha(1,1,0) + \beta(0,1,1) + \gamma(1,0,1) = v$ for some $\alpha, \beta, \gamma \in \mathbb{R}$ and $v \in \mathbb{R}^3$, then choose the set of correct options. ***1 point***

- [ ] If $v = (0,0,0)$, then one of the possible solutions of $\alpha, \beta,$ and γ is $\alpha = \frac{1}{2} = \gamma$, and $\beta = -\frac{1}{2}$.
- [ ] If $v = (0,0,0)$, then one of the possible solutions of $\alpha, \beta,$ and γ is $\alpha = \beta = \gamma = 0$.
- [ ] If $v = (0,0,0)$, then the possible solution of $\alpha, \beta,$ and γ is unique.
- [ ] If $v = (1,0,0)$, then one of the possible solutions of $\alpha, \beta,$ and γ is $\alpha = \frac{1}{2} = \gamma$, and $\beta = -\frac{1}{2}$.
- [ ] The set $\{(1,1,0),(0,1,1),(1,0,1)\}$ is linearly independent.

No, the answer is incorrect.
Score: 0

Accepted Answers:

If $v = (0, 0, 0)$, then one of the possible solutions of $\alpha, \beta,$ and γ is $\alpha = \beta = \gamma = 0$.
If $v = (0, 0, 0)$, then the possible solution of $\alpha, \beta,$ and γ is unique.
If $v = (1, 0, 0)$, then one of the possible solutions of $\alpha, \beta,$ and γ is $\alpha = \frac{1}{2} = \gamma$, and $\beta = -\frac{1}{2}$.
The set $\{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}$ is linearly independent.

3) Consider a set of vectors $S = \{(1, 2), (-1, 2), (3, 1), (-3, 1)\}$ in $\mathbb{R}^2$. Choose the set of correct options. *1 point*

- [ ] $\{(1, 2), (-1, 2)\}$ is a linearly dependent set.
- [ ] $\{(-1, 2), (-3, 1)\}$ is a linearly independent set.
- [ ] $\{(-3, 1), (-1, 2), (3, 1)\}$ is a linearly dependent set.
- [ ] $\{(3, 1), (-3, 1)\}$ is a linearly dependent set.
- [ ] $\{(3, 1), (1, 2), (-3, 1)\}$ is a linearly dependent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(-1, 2), (-3, 1)\}$ is a linearly independent set.
$\{(-3, 1), (-1, 2), (3, 1)\}$ is a linearly dependent set.
$\{(3, 1), (1, 2), (-3, 1)\}$ is a linearly dependent set.

4) Let V be a vector space and non zero vectors $v_1, v_2, v_3,$ and $v_4 \in V$. If v_1 is a linear combination of v_i, for $i = 2, 3, 4$, then which of the following is (are) correct? *1 point*

- [ ] The set $\{v_1, v_2, v_3, v_4\}$ cannot be linearly independent.
- [ ] The set $\{v_2, v_3, v_4\}$ must be linearly dependent.
- [ ] The set $\{v_2, v_3, v_4\}$ must be linearly independent.
- [ ] The set $\{v_1\}$ must be linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set $\{v_1, v_2, v_3, v_4\}$ cannot be linearly independent.
The set $\{v_1\}$ must be linearly independent.

5) Choose the correct set of options. ***1 point***

- [ ] Union of two distinct linearly independent sets may be linearly independent.
- [ ] Union of two distinct linearly independent sets must be linearly dependent.
- [ ] Non-empty intersection of two linearly independent sets must be linearly independent.
- [ ] Non-empty intersection of two linearly independent sets must be linearly dependent.

No, the answer is incorrect.
Score: 0

Accepted Answers:

Union of two distinct linearly independent sets may be linearly independent.
Non-empty intersection of two linearly independent sets must be linearly independent.

6) If vectors v_1, v_2, v_3 are linearly independent and $\alpha v_1 + \beta v_2 + \gamma v_3 = 0$, then find the value of $\alpha + \beta + \gamma$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

7) Consider the set of matrices $S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \begin{bmatrix} 7 \\ 8 \\ x \end{bmatrix} \right\} \subset M_{3\times1}(\mathbb{R})$. If $x \in \{9, 4, 2, 0\}$, then find the maximum value of x so that S is linearly independent.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 4

1 point

Level 2

8) Let S be a linearly independent subset of $\mathbb{R}^3$. Choose the correct set of options. ***1 point***

- [ ] $S \cup \{(1, 0, 0)\}$ must be linearly independent.
- [ ] $S \cup \{v\}$ must be linearly independent for any $v \in \mathbb{R}^3$.
- [ ] $S \setminus \{v\}$ must be linearly independent for any $v \in \mathbb{R}^3$.
- [ ] There may exist some $v \in \mathbb{R}^3$, such that $S \cup \{v\}$ is still linearly independent.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$S \setminus \{v\}$ must be linearly independent for any $v \in \mathbb{R}^3$.
There may exist some $v \in \mathbb{R}^3$, such that $S \cup \{v\}$ is still linearly independent.

9) Consider a set of vectors $S = \{(1, 1, -1), (-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2), (1, 0, -2)\}$. Choose the set of correct options. ***1 point***

- [ ] The singleton set $\{(0, 1, -2)\}$ is linearly dependent.
- [ ] If $\alpha, \beta \in S$ and α, β are distinct then $\{\alpha, \beta\}$ is a linearly independent set of vectors.
- [ ] The set $\{(-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2)\}$ is a linearly independent set of vectors.
- [ ] The set S is a linearly independent set of vectors.
- [ ] The set $\{\alpha, \beta, \gamma\}$ is a linearly dependent set of vectors for any $\alpha, \beta, \gamma \in S$, where all the three are distinct vectors.
- [ ] The set $\{\alpha, \beta, \gamma, \delta\}$ is a linearly independent set of vectors for any $\alpha, \beta, \gamma, \delta \in S$, where all the four are distinct vectors.

No, the answer is incorrect.
Score: 0

Accepted Answers:

If $\alpha, \beta \in S$ and α, β are distinct then $\{\alpha, \beta\}$ is a linearly independent set of vectors.
The set $\{(-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2)\}$ is a linearly independent set of vectors.

10) Let S be the solution set of a system of homogeneous linear equations with 3 variables and 3 equations, whose matrix representation is as follows: ***1 point***

$$Ax = 0$$

where A is the 3×3 coefficient matrix and x denotes the column vector $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$. Choose the set of correct options.

- [ ] If v_1 and v_2 are in S, then any linear combination of v_1 and v_2 will also be in S.
- [ ] The set S will be a subspace of R^3, with respect to usual addition and scalar multiplication as in R^3.
- [ ] The set $\{v_1, v_2, v_1 - v_2\}$ is a linearly dependent subset in S.
- [ ] The set $\{v_1, v_2\}$ is a linearly independent subset in S if v_1 is not a scalar multiple of v_2.

No, the answer is incorrect.
Score: 0

Accepted Answers:

If v_1 and v_2 are in S, then any linear combination of v_1 and v_2 will also be in S.
The set S will be a subspace of R^3, with respect to usual addition and scalar multiplication as in R^3.
The set $\{v_1, v_2, v_1 - v_2\}$ is a linearly dependent subset in S.
The set $\{v_1, v_2\}$ is a linearly independent subset in S if v_1 is not a scalar multiple of v_2.

Your score is: 0/10